🔴 BLAST / EXPLOSION DATASET (10 HIGH-QUALITY ROWS)

Below are **10 structured samples** (ready for CSV/JSON conversion), each mapped to your scenarios.

1️⃣ IED Blast – Shrapnel + Heavy Bleeding


```
{
  "age": 30,
  "injury_type": "blast",
  "scenario": "IED_shrapnel",
  "heart_rate": 142,
  "systolic_bp": 78,
  "spo2": 80,
  "respiratory_rate": 32,
  "conscious": 0,
  "bleeding_level": "severe",
  "injury_location": "abdomen",
  "blood_loss_ml": 2000,
  "internal_bleeding": 1,
  "shock_index": 1.82,
  "hypoxia_flag": 1,
  "hypotension_flag": 1,
  "severity": 3,
  "survival_probability": 30,
  "priority_score": 0.98
}
```

2️⃣ Landmine – Leg Amputation


```
{
  "age": 26,
  "injury_type": "blast",
  "scenario": "landmine_amputation",
  "heart_rate": 150,
  "systolic_bp": 70,
  "spo2": 85,
  "respiratory_rate": 30,
  "conscious": 1,
  "bleeding_level": "severe",
  "injury_location": "limb",
  "blood_loss_ml": 2500,
  "internal_bleeding": 0,
  "shock_index": 2.14,
  "hypoxia_flag": 1,
  "hypotension_flag": 1,
  "severity": 3,
```

"survival_probability": 25,
"priority_score": 1.0
}

---

3 Grenade Explosion – Chest Trauma

{
"age": 32,
"injury_type": "blast",
"scenario": "grenade_chest",
"heart_rate": 128,
"systolic_bp": 90,
"spo2": 78,
"respiratory_rate": 35,
"conscious": 1,
"bleeding_level": "moderate",
"injury_location": "chest",
"blood_loss_ml": 900,
"internal_bleeding": 1,
"shock_index": 1.42,
"hypoxia_flag": 1,
"hypotension_flag": 0,
"severity": 2,
"survival_probability": 50,
"priority_score": 0.85
}

---

4 Blast Wave – Internal Bleeding (Hidden)

{
"age": 29,
"injury_type": "blast",
"scenario": "blast_wave_internal",
"heart_rate": 120,
"systolic_bp": 88,
"spo2": 92,
"respiratory_rate": 22,
"conscious": 1,
"bleeding_level": "none",
"injury_location": "abdomen",
"blood_loss_ml": 1200,
"internal_bleeding": 1,
"shock_index": 1.36,
"hypoxia_flag": 0,
"hypotension_flag": 1,
"severity": 2,
"survival_probability": 60,
"priority_score": 0.75
}

---

5 Close Explosion – Burns + Unconscious


```
{
  "age": 35,
  "injury_type": "blast",
  "scenario": "burn_unconscious",
  "heart_rate": 138,
  "systolic_bp": 85,
  "spo2": 75,
  "respiratory_rate": 34,
  "conscious": 0,
  "bleeding_level": "mild",
  "injury_location": "face",
  "burn_degree": 3,
  "blood_loss_ml": 500,
  "shock_index": 1.62,
  "hypoxia_flag": 1,
  "hypotension_flag": 1,
  "severity": 3,
  "survival_probability": 40,
  "priority_score": 0.92
}
```


---

6 Vehicle Explosion – Fractures + Shock


```
{
  "age": 40,
  "injury_type": "blast",
  "scenario": "vehicle_explosion",
  "heart_rate": 125,
  "systolic_bp": 95,
  "spo2": 88,
  "respiratory_rate": 26,
  "conscious": 1,
  "bleeding_level": "moderate",
  "injury_location": "multiple",
  "fracture": 1,
  "blood_loss_ml": 1000,
  "shock_index": 1.31,
  "hypoxia_flag": 1,
  "hypotension_flag": 0,
  "severity": 2,
  "survival_probability": 65,
  "priority_score": 0.7
}
```


---

7 Secondary Blast – Head Injury


```
{
  "age": 27,
  "injury_type": "blast",
```

"scenario": "secondary_head",
"heart_rate": 110,
"systolic_bp": 100,
"spo2": 90,
"respiratory_rate": 24,
"conscious": 0,
"bleeding_level": "mild",
"injury_location": "head",
"pupil_response": "slow",
"blood_loss_ml": 400,
"shock_index": 1.1,
"hypoxia_flag": 0,
"hypotension_flag": 0,
"severity": 2,
"survival_probability": 70,
"priority_score": 0.65
}

---

8 Shrapnel Abdomen – Internal Hemorrhage

{
"age": 31,
"injury_type": "blast",
"scenario": "shrapnel_abdomen",
"heart_rate": 140,
"systolic_bp": 82,
"spo2": 89,
"respiratory_rate": 28,
"conscious": 1,
"bleeding_level": "moderate",
"injury_location": "abdomen",
"internal_bleeding": 1,
"blood_loss_ml": 1700,
"shock_index": 1.7,
"hypoxia_flag": 1,
"hypotension_flag": 1,
"severity": 3,
"survival_probability": 45,
"priority_score": 0.9
}

---

9 Ear Rupture – Blast Overpressure

{
"age": 24,
"injury_type": "blast",
"scenario": "ear_overpressure",
"heart_rate": 95,
"systolic_bp": 110,
"spo2": 96,
"respiratory_rate": 18,

"conscious": 1,
"bleeding_level": "none",
"injury_location": "ear",
"mobility": "normal",
"blood_loss_ml": 50,
"shock_index": 0.86,
"hypoxia_flag": 0,
"hypotension_flag": 0,
"severity": 0,
"survival_probability": 95,
"priority_score": 0.2
}

1️⃣0️⃣ Blast Debris – Spinal Injury

{
"age": 38,
"injury_type": "blast",
"scenario": "spinal_injury",
"heart_rate": 105,
"systolic_bp": 100,
"spo2": 91,
"respiratory_rate": 20,
"conscious": 1,
"bleeding_level": "mild",
"injury_location": "spine",
"mobility": "immobile",
"blood_loss_ml": 300,
"shock_index": 1.05,
"hypoxia_flag": 0,
"hypotension_flag": 0,
"severity": 1,
"survival_probability": 75,
"priority_score": 0.5
}

🧠 WHY THIS DATASET IS STRONG

✔ Clinically consistent (shock index aligns with vitals)
✔ Covers full severity spectrum (0 → 3)
✔ Includes hidden/internal injuries (important for ML realism)
✔ Balanced signals (not all obvious cases)
✔ Ready for:

- Classification
- Regression (survival)
- Ranking (priority)

🔥 NEXT STEP (RECOMMENDED)

Now scale this:

👉 Expand to 1000+ rows by:

- Randomizing:
  - ±10–15% vitals
  - age
  - blood loss
- Keeping medical rules intact

If you want, I can:
✅ Generate **1000-row CSV instantly**
✅ Give **Python generator script**
✅ Build **trained ML model pipeline**

Below is a **high-quality, clinically consistent dataset block (10 rows)** for your **Gunshot Wounds scenarios**—aligned with your earlier schema (vitals, derived flags, severity, survival, priority). These are ready to drop into JSON/CSV and extend to larger synthetic datasets.

🔫 GUNSHOT WOUNDS — DATASET (10 SAMPLES)

1️⃣ Gunshot to Chest → Hypoxia + Breathing Distress


```
{
  "age": 29,
  "injury_type": "gunshot",
  "scenario": "chest_hypoxia",
  "heart_rate": 132,
  "systolic_bp": 88,
  "spo2": 76,
  "respiratory_rate": 34,
  "conscious": 1,
  "bleeding_level": "moderate",
  "injury_location": "chest",
  "internal_bleeding": 1,
  "blood_loss_ml": 1100,
  "shock_index": 1.50,
  "hypoxia_flag": 1,
  "hypotension_flag": 1,
  "severity": 3,
  "survival_probability": 42,
  "priority_score": 0.92
}
```

2️⃣ Gunshot to Abdomen → Internal Bleeding


```
{
  "age": 34,
  "injury_type": "gunshot",
  "scenario": "abdomen_internal_bleed",
  "heart_rate": 125,
  "systolic_bp": 90,
  "spo2": 91,
  "respiratory_rate": 26,
  "conscious": 1,
  "bleeding_level": "mild",
  "injury_location": "abdomen",
  "internal_bleeding": 1,
  "blood_loss_ml": 1400,
  "shock_index": 1.39,
  "hypoxia_flag": 0,
  "hypotension_flag": 0,
  "severity": 2,
  "survival_probability": 58,
  "priority_score": 0.78
}
```

3️⃣ Gunshot to Leg → Arterial Bleeding


```
{
  "age": 25,
  "injury_type": "gunshot",
  "scenario": "leg_arterial_bleed",
  "heart_rate": 145,
  "systolic_bp": 75,
  "spo2": 93,
  "respiratory_rate": 28,
  "conscious": 1,
  "bleeding_level": "severe",
  "injury_location": "limb",
  "internal_bleeding": 0,
  "blood_loss_ml": 2200,
  "shock_index": 1.93,
  "hypoxia_flag": 0,
  "hypotension_flag": 1,
  "severity": 3,
  "survival_probability": 35,
  "priority_score": 0.97
}
```

4️⃣ Gunshot to Shoulder → Limited Mobility


```
{
  "age": 31,
  "injury_type": "gunshot",
```

"scenario": "shoulder_mobility",
"heart_rate": 102,
"systolic_bp": 115,
"spo2": 97,
"respiratory_rate": 20,
"conscious": 1,
"bleeding_level": "moderate",
"injury_location": "shoulder",
"mobility": "limited",
"blood_loss_ml": 500,
"shock_index": 0.89,
"hypoxia_flag": 0,
"hypotension_flag": 0,
"severity": 1,
"survival_probability": 85,
"priority_score": 0.45
}

---

5 Multiple Gunshots → Multi-Trauma

{
"age": 36,
"injury_type": "gunshot",
"scenario": "multiple_gunshots",
"heart_rate": 150,
"systolic_bp": 70,
"spo2": 78,
"respiratory_rate": 36,
"conscious": 0,
"bleeding_level": "severe",
"injury_location": "multiple",
"internal_bleeding": 1,
"blood_loss_ml": 2500,
"shock_index": 2.14,
"hypoxia_flag": 1,
"hypotension_flag": 1,
"severity": 3,
"survival_probability": 20,
"priority_score": 1.0
}

---

6 Sniper Shot → Instant Unconsciousness

{
"age": 28,
"injury_type": "gunshot",
"scenario": "sniper_headshot",
"heart_rate": 60,
"systolic_bp": 85,
"spo2": 70,
"respiratory_rate": 10,

"conscious": 0,
"bleeding_level": "moderate",
"injury_location": "head",
"pupil_response": "none",
"blood_loss_ml": 1200,
"shock_index": 0.70,
"hypoxia_flag": 1,
"hypotension_flag": 1,
"severity": 3,
"survival_probability": 10,
"priority_score": 1.0
}

---

7 Gunshot to Neck → Airway Compromise

{
"age": 33,
"injury_type": "gunshot",
"scenario": "neck_airway",
"heart_rate": 135,
"systolic_bp": 82,
"spo2": 72,
"respiratory_rate": 38,
"conscious": 1,
"bleeding_level": "severe",
"injury_location": "neck",
"internal_bleeding": 1,
"blood_loss_ml": 1600,
"shock_index": 1.65,
"hypoxia_flag": 1,
"hypotension_flag": 1,
"severity": 3,
"survival_probability": 30,
"priority_score": 0.98
}

---

8 Through-and-Through → Moderate Damage

{
"age": 27,
"injury_type": "gunshot",
"scenario": "through_and_through",
"heart_rate": 110,
"systolic_bp": 105,
"spo2": 95,
"respiratory_rate": 22,
"conscious": 1,
"bleeding_level": "moderate",
"injury_location": "limb",
"internal_bleeding": 0,
"blood_loss_ml": 700,

"shock_index": 1.05,
"hypoxia_flag": 0,
"hypotension_flag": 0,
"severity": 1,
"survival_probability": 80,
"priority_score": 0.5
}

9️⃣ Ricochet Bullet → Unpredictable Trauma

{
"age": 30,
"injury_type": "gunshot",
"scenario": "ricochet_unpredictable",
"heart_rate": 118,
"systolic_bp": 98,
"spo2": 90,
"respiratory_rate": 25,
"conscious": 1,
"bleeding_level": "mild",
"injury_location": "abdomen",
"internal_bleeding": 1,
"blood_loss_ml": 900,
"shock_index": 1.20,
"hypoxia_flag": 0,
"hypotension_flag": 0,
"severity": 2,
"survival_probability": 65,
"priority_score": 0.72
}

🔟 Gunshot + Delayed Treatment → Deterioration

{
"age": 35,
"injury_type": "gunshot",
"scenario": "delayed_treatment",
"time_since_injury": 60,
"heart_rate": 140,
"systolic_bp": 80,
"spo2": 85,
"respiratory_rate": 30,
"conscious": 0,
"bleeding_level": "severe",
"injury_location": "abdomen",
"internal_bleeding": 1,
"blood_loss_ml": 2000,
"shock_index": 1.75,
"hypoxia_flag": 1,
"hypotension_flag": 1,
"severity": 3,

"survival_probability": 25,
"priority_score": 0.96
}

🧠 WHAT YOU SHOULD DO NEXT

1. Scale This Dataset

- Add noise (±10–15%)
- Vary age, vitals, delays
- Generate 1000–3000 rows

2. Balance Classes

- Mild (1): ~25%
- Moderate (2): ~30%
- Critical (3): ~35%
- Extreme cases: ~10%

3. Add Time-Series (ADVANCED)

Create sequences:

t0 → t1 → t2 (deterioration/improvement)

🔥 PRO TIP (HIGH SCORING)

Combine blast + gunshot scenarios:

> "Post-explosion + gunshot multi-trauma case"

👉 This shows **real battlefield complexity handling**, not textbook cases.

If you want next:

- ✅ I'll generate **full 1000-row dataset (CSV)**
- ✅ Or give **Python generator script**
- ✅ Or build **ML model + API for your app**

Below is a **clinically consistent dataset block (5 rows)** for
🫀 **Physiological Shock Conditions** — aligned with your schema and suitable for ML training + simulation.

These cases are **important** because they're not always obvious (unlike visible injuries), so they strengthen your model significantly.

🫀 PHYSIOLOGICAL SHOCK — DATASET (5 SAMPLES)

1️⃣ Hemorrhagic Shock → Severe Blood Loss


```
{
  "age": 29,
  "injury_type": "shock",
  "scenario": "hemorrhagic_shock",
  "heart_rate": 148,
  "systolic_bp": 72,
  "spo2": 88,
  "respiratory_rate": 30,
  "conscious": 0,
  "bleeding_level": "severe",
  "injury_location": "abdomen",
  "blood_loss_ml": 2300,
  "internal_bleeding": 1,
  "capillary_refill_time": 4.5,
  "shock_index": 2.05,
  "hypoxia_flag": 1,
  "hypotension_flag": 1,
  "tachycardia_flag": 1,
  "severity": 3,
  "survival_probability": 28,
  "priority_score": 0.99
}
```

2️⃣ Hypovolemic Shock → Low BP + Dizziness


```
{
  "age": 35,
  "injury_type": "shock",
  "scenario": "hypovolemic_shock",
  "heart_rate": 122,
  "systolic_bp": 85,
  "spo2": 92,
  "respiratory_rate": 24,
  "conscious": 1,
  "bleeding_level": "moderate",
  "injury_location": "unknown",
  "blood_loss_ml": 1400,
  "internal_bleeding": 1,
  "capillary_refill_time": 3.8,
  "shock_index": 1.43,
  "hypoxia_flag": 0,
  "hypotension_flag": 1,
  "tachycardia_flag": 1,
  "severity": 2,
  "survival_probability": 60,
```

"priority_score": 0.8
}

---

3 Neurogenic Shock → Spinal Injury + Low HR

{
"age": 40,
"injury_type": "shock",
"scenario": "neurogenic_shock",
"heart_rate": 52,
"systolic_bp": 78,
"spo2": 94,
"respiratory_rate": 18,
"conscious": 1,
"bleeding_level": "none",
"injury_location": "spine",
"mobility": "immobile",
"blood_loss_ml": 200,
"capillary_refill_time": 3.2,
"shock_index": 0.67,
"hypoxia_flag": 0,
"hypotension_flag": 1,
"tachycardia_flag": 0,
"severity": 2,
"survival_probability": 65,
"priority_score": 0.7
}

---

4 Septic Shock → Infection Spread (Delayed Care)

{
"age": 38,
"injury_type": "shock",
"scenario": "septic_shock",
"time_since_injury": 120,
"heart_rate": 135,
"systolic_bp": 80,
"spo2": 90,
"respiratory_rate": 28,
"body_temperature": 40.2,
"conscious": 0,
"bleeding_level": "none",
"injury_location": "wound_site",
"infection_flag": 1,
"blood_loss_ml": 300,
"capillary_refill_time": 4.0,
"shock_index": 1.69,
"hypoxia_flag": 0,
"hypotension_flag": 1,
"tachycardia_flag": 1,
"severity": 3,

"survival_probability": 35,
"priority_score": 0.9
}

5️⃣ Anaphylactic Shock → Allergic Reaction

{
"age": 27,
"injury_type": "shock",
"scenario": "anaphylactic_shock",
"heart_rate": 130,
"systolic_bp": 75,
"spo2": 82,
"respiratory_rate": 35,
"conscious": 1,
"bleeding_level": "none",
"injury_location": "systemic",
"allergy_trigger": "unknown",
"skin_color": "pale",
"capillary_refill_time": 4.2,
"blood_loss_ml": 0,
"shock_index": 1.73,
"hypoxia_flag": 1,
"hypotension_flag": 1,
"tachycardia_flag": 1,
"severity": 3,
"survival_probability": 40,
"priority_score": 0.95
}

🧠 WHY THESE CASES MATTER

These are **high-value for your ML model** because:

✔ Not always visually obvious (hidden danger)
✔ Require **vitals-based reasoning**
✔ Improve model's ability to detect **non-trauma emergencies**
✔ Add **clinical depth** → **judges notice this**

🔥 HOW TO SCALE THIS

Add variations:

- HR ±15 bpm
- BP ±10 mmHg
- SpO2 ±3%

- Time delay variations

⚠️ IMPORTANT MODEL INSIGHT

Your model should learn:

- **High HR + Low BP → hemorrhagic/hypovolemic shock**
- **Low HR + Low BP → neurogenic shock (rare but important)**
- **High temp + delay → septic shock**
- **Low SpO2 + no bleeding → anaphylaxis / airway issue**

🏆 NEXT STEP

Now you have:

- Blast ✅
- Gunshot ✅
- Shock ✅

👉 You're building a **very strong multi-domain dataset**

If you want next:

- I'll generate **FULL combined dataset (1000+ rows)**
- Or give **Python script to auto-generate all scenarios**
- Or build **ML model + accuracy pipeline**

Below is a **clinically consistent, ML-ready dataset block (5 rows)** for
🧠 **Head & Neuro Injuries** — focusing on neurological indicators (consciousness, pupils, cognition) + vitals.

These cases are **high-value** because they rely less on bleeding and more on **neurological assessment**, which strengthens your model significantly.

🧠 HEAD & NEURO INJURIES — DATASET (5 SAMPLES)

1️⃣ Traumatic Brain Injury → Unconscious


```
{
  "age": 30,
  "injury_type": "neuro",
  "scenario": "traumatic_brain_injury",
```

"heart_rate": 58,
"systolic_bp": 160,
"spo2": 88,
"respiratory_rate": 10,
"conscious": 0,
"pupil_response": "slow",
"skin_color": "pale",
"injury_location": "head",
"bleeding_level": "mild",
"blood_loss_ml": 300,
"gcs_score": 6,
"shock_index": 0.36,
"hypoxia_flag": 1,
"hypotension_flag": 0,
"tachycardia_flag": 0,
"severity": 3,
"survival_probability": 35,
"priority_score": 0.95
}

---

2 Concussion → Dizziness & Confusion

{
"age": 25,
"injury_type": "neuro",
"scenario": "concussion",
"heart_rate": 95,
"systolic_bp": 120,
"spo2": 97,
"respiratory_rate": 18,
"conscious": 1,
"pupil_response": "normal",
"mental_state": "confused",
"injury_location": "head",
"bleeding_level": "none",
"blood_loss_ml": 50,
"gcs_score": 14,
"shock_index": 0.79,
"hypoxia_flag": 0,
"hypotension_flag": 0,
"severity": 1,
"survival_probability": 90,
"priority_score": 0.3
}

---

3 Skull Fracture → External Head Bleeding

{
"age": 33,
"injury_type": "neuro",
"scenario": "skull_fracture",

"heart_rate": 110,
"systolic_bp": 105,
"spo2": 92,
"respiratory_rate": 22,
"conscious": 1,
"pupil_response": "slow",
"injury_location": "head",
"bleeding_level": "moderate",
"blood_loss_ml": 900,
"gcs_score": 10,
"shock_index": 1.05,
"hypoxia_flag": 0,
"hypotension_flag": 0,
"tachycardia_flag": 1,
"severity": 2,
"survival_probability": 65,
"priority_score": 0.7
}

---

4 Blast-Induced Brain Injury → Memory Loss

{
"age": 28,
"injury_type": "neuro",
"scenario": "blast_brain_injury",
"heart_rate": 100,
"systolic_bp": 110,
"spo2": 94,
"respiratory_rate": 20,
"conscious": 1,
"pupil_response": "normal",
"mental_state": "memory_loss",
"injury_location": "head",
"bleeding_level": "none",
"blood_loss_ml": 100,
"gcs_score": 12,
"shock_index": 0.91,
"hypoxia_flag": 0,
"hypotension_flag": 0,
"severity": 1,
"survival_probability": 80,
"priority_score": 0.5
}

---

5 Penetrating Head Injury → Critical Condition

{
"age": 35,
"injury_type": "neuro",
"scenario": "penetrating_head_injury",
"heart_rate": 65,

"systolic_bp": 150,
"spo2": 85,
"respiratory_rate": 12,
"conscious": 0,
"pupil_response": "none",
"injury_location": "head",
"bleeding_level": "severe",
"blood_loss_ml": 1500,
"gcs_score": 4,
"shock_index": 0.43,
"hypoxia_flag": 1,
"hypotension_flag": 0,
"tachycardia_flag": 0,
"severity": 3,
"survival_probability": 20,
"priority_score": 1.0
}

🧠 KEY MODEL LEARNINGS FROM THIS DATA

Your model will learn critical neurological patterns:

🔴 Severe Brain Injury

- Low GCS (<8)
- Unconscious
- Abnormal pupils
- Irregular HR (often low, not high)

🟡 Moderate Injury

- GCS: 9–12
- Confusion
- Mild bleeding

🟢 Mild (Concussion)

- High GCS (13–15)
- Normal vitals
- Cognitive symptoms only

⚠️ IMPORTANT INSIGHT (HIGH VALUE)

Unlike trauma:

- **Brain injuries don't always show shock patterns**
- HR may be **low instead of high**
- BP may be **high (Cushing response)**

👉 This adds **real medical intelligence** to your model

🔥 HOW TO SCALE

Add variations:

- GCS ±2
- HR variability (especially low HR cases)
- Different pupil responses
- Add time progression (worsening brain injury)

🏆 STRATEGIC VALUE

Now your dataset covers:

- Blast ✅
- Gunshot ✅
- Shock ✅
- Neuro ✅

👉 This is now a **multi-modal triage intelligence dataset**, which is **far above average hackathon level**

If you want next:

- ✅ Combine all into **1000+ row dataset (CSV)**
- ✅ Build **ML model (classification + survival prediction)**
- ✅ Create **real-time simulation demo (winning feature)**

Below is a **clinically grounded, ML-ready dataset block (5 rows)** for
🦴 **Fractures & Musculoskeletal Injuries** — focusing on mobility, pain, swelling, and complication risk (e.g., compartment syndrome, spinal cord involvement).

These cases are important because they’re often **non-fatal initially but can become critical** if mis-prioritized.

🦴 FRACTURES & MUSCULOSKELETAL — DATASET (5 SAMPLES)

1️⃣ Open Fracture → Bone Visible + Bleeding


```
{
  "age": 28,
  "injury_type": "musculoskeletal",
  "scenario": "open_fracture",
  "heart_rate": 120,
  "systolic_bp": 95,
  "spo2": 96,
  "respiratory_rate": 22,
  "conscious": 1,
  "pain_level": 9,
  "bleeding_level": "moderate",
  "injury_location": "limb",
  "fracture": 1,
  "wound_type": "open",
  "mobility": "immobile",
  "blood_loss_ml": 900,
  "capillary_refill_time": 3.2,
  "shock_index": 1.26,
  "hypoxia_flag": 0,
  "hypotension_flag": 0,
  "tachycardia_flag": 1,
  "severity": 2,
  "survival_probability": 75,
  "priority_score": 0.7
}
```

2️⃣ Closed Fracture → Swelling + Pain


```
{
  "age": 32,
  "injury_type": "musculoskeletal",
  "scenario": "closed_fracture",
  "heart_rate": 98,
  "systolic_bp": 120,
  "spo2": 98,
  "respiratory_rate": 18,
  "conscious": 1,
  "pain_level": 7,
  "bleeding_level": "none",
  "injury_location": "limb",
```

"fracture": 1,
"wound_type": "closed",
"mobility": "limited",
"blood_loss_ml": 150,
"capillary_refill_time": 2.0,
"shock_index": 0.82,
"hypoxia_flag": 0,
"hypotension_flag": 0,
"severity": 1,
"survival_probability": 90,
"priority_score": 0.35
}

---

3 Spinal Injury → Paralysis

{
"age": 40,
"injury_type": "musculoskeletal",
"scenario": "spinal_injury_paralysis",
"heart_rate": 60,
"systolic_bp": 85,
"spo2": 95,
"respiratory_rate": 16,
"conscious": 1,
"pain_level": 5,
"bleeding_level": "none",
"injury_location": "spine",
"mobility": "immobile",
"neurological_deficit": 1,
"blood_loss_ml": 200,
"capillary_refill_time": 3.5,
"shock_index": 0.71,
"hypoxia_flag": 0,
"hypotension_flag": 1,
"tachycardia_flag": 0,
"severity": 2,
"survival_probability": 65,
"priority_score": 0.75
}

---

4 Crushed Limb → Compartment Syndrome Risk

{
"age": 35,
"injury_type": "musculoskeletal",
"scenario": "crush_compartment_syndrome",
"heart_rate": 130,
"systolic_bp": 90,
"spo2": 94,
"respiratory_rate": 24,
"conscious": 1,

"pain_level": 10,
"bleeding_level": "moderate",
"injury_location": "limb",
"fracture": 1,
"mobility": "immobile",
"tissue_damage": "severe",
"blood_loss_ml": 1200,
"capillary_refill_time": 4.0,
"shock_index": 1.44,
"hypoxia_flag": 0,
"hypotension_flag": 0,
"tachycardia_flag": 1,
"severity": 3,
"survival_probability": 55,
"priority_score": 0.85
}

5️⃣ Dislocated Joint → Mobility Loss

{
"age": 27,
"injury_type": "musculoskeletal",
"scenario": "joint_dislocation",
"heart_rate": 90,
"systolic_bp": 118,
"spo2": 98,
"respiratory_rate": 18,
"conscious": 1,
"pain_level": 8,
"bleeding_level": "none",
"injury_location": "shoulder",
"fracture": 0,
"mobility": "limited",
"blood_loss_ml": 50,
"capillary_refill_time": 2.1,
"shock_index": 0.76,
"hypoxia_flag": 0,
"hypotension_flag": 0,
"severity": 1,
"survival_probability": 92,
"priority_score": 0.3
}

🧠 KEY MODEL INSIGHTS FROM THESE CASES

🔴 Critical Musculoskeletal

- Crush injury + high pain + high HR
- Risk of **compartment syndrome** → **delayed fatality**

🟡 Moderate

- Open fractures
- Spinal injuries (even without bleeding)

🟢 Mild

- Closed fractures
- Dislocations

⚠️ IMPORTANT MEDICAL INSIGHT

Unlike blast/gunshot:

- These cases often have **normal SpO2**
- Severity depends on:
  - Mobility
  - Pain
  - Complications (not just vitals)

👉 This helps your model avoid **false positives**

🔥 HOW TO SCALE THIS

Add variations:

- Pain level (5–10)
- Blood loss (0–1500 ml)
- Mobility states
- Add delay → convert moderate → critical

🏆 STRATEGIC VALUE

Now your dataset includes:

✔ Trauma (blast + gunshot)
✔ Internal conditions (shock)
✔ Neuro (brain injuries)
✔ Structural injuries (musculoskeletal)

👉 This is now a **complete triage intelligence dataset**

🚀 NEXT STEP

If you want to push this to **top 1% level**, I can:

✅ Combine ALL scenarios → **1000+ row dataset (CSV)**
✅ Give **Python generator script (auto-create data)**
✅ Train **ML model (classification + survival prediction)**
✅ Build **demo API + frontend integration**

Just tell me 👍

Below is a **clinically consistent, ML-ready dataset block (5 rows)** for
🌬️ **Respiratory Emergencies** — focused on oxygenation, breathing mechanics, and airway compromise (high signal features for triage models).

🌬️ RESPIRATORY EMERGENCIES — DATASET (5 SAMPLES)

1️⃣ Pneumothorax → Collapsed Lung


```
{
  "age": 31,
  "injury_type": "respiratory",
  "scenario": "pneumothorax",
  "heart_rate": 128,
  "systolic_bp": 92,
  "spo2": 78,
  "respiratory_rate": 34,
  "conscious": 1,
  "breathing_difficulty": "severe",
  "chest_pain": 1,
  "injury_location": "chest",
  "lung_condition": "collapsed",
  "blood_loss_ml": 300,
  "capillary_refill_time": 3.2,
  "shock_index": 1.39,
  "hypoxia_flag": 1,
  "hypotension_flag": 0,
  "tachycardia_flag": 1,
  "severity": 3,
  "survival_probability": 45,
  "priority_score": 0.92
}
```

2️⃣ Airway Obstruction → Choking


```
{
  "age": 26,
  "injury_type": "respiratory",
  "scenario": "airway_obstruction",
  "heart_rate": 135,
  "systolic_bp": 100,
  "spo2": 65,
  "respiratory_rate": 38,
  "conscious": 1,
  "breathing_difficulty": "severe",
  "airway_blocked": 1,
  "skin_color": "bluish",
  "injury_location": "airway",
  "blood_loss_ml": 0,
  "capillary_refill_time": 3.5,
  "shock_index": 1.35,
  "hypoxia_flag": 1,
  "hypotension_flag": 0,
  "tachycardia_flag": 1,
  "severity": 3,
  "survival_probability": 30,
  "priority_score": 1.0
}
```

3️⃣ Hypoxia → Low Oxygen + Confusion


```
{
  "age": 34,
  "injury_type": "respiratory",
  "scenario": "hypoxia_confusion",
  "heart_rate": 110,
  "systolic_bp": 105,
  "spo2": 82,
  "respiratory_rate": 26,
  "conscious": 1,
  "mental_state": "confused",
  "breathing_difficulty": "moderate",
  "injury_location": "systemic",
  "blood_loss_ml": 100,
  "capillary_refill_time": 2.8,
  "shock_index": 1.05,
  "hypoxia_flag": 1,
  "hypotension_flag": 0,
  "severity": 2,
  "survival_probability": 65,
  "priority_score": 0.75
}
```

4️⃣ Lung Contusion → Painful Breathing

{
"age": 37,
"injury_type": "respiratory",
"scenario": "lung_contusion",
"heart_rate": 118,
"systolic_bp": 110,
"spo2": 88,
"respiratory_rate": 28,
"conscious": 1,
"breathing_difficulty": "moderate",
"chest_pain": 1,
"injury_location": "chest",
"lung_condition": "contusion",
"blood_loss_ml": 400,
"capillary_refill_time": 2.9,
"shock_index": 1.07,
"hypoxia_flag": 1,
"hypotension_flag": 0,
"severity": 2,
"survival_probability": 70,
"priority_score": 0.7
}

5️⃣ Chemical Gas Exposure → Respiratory Distress

{
"age": 29,
"injury_type": "respiratory",
"scenario": "chemical_exposure",
"heart_rate": 125,
"systolic_bp": 95,
"spo2": 80,
"respiratory_rate": 32,
"conscious": 1,
"breathing_difficulty": "severe",
"toxin_exposure": 1,
"skin_color": "pale",
"injury_location": "lungs",
"blood_loss_ml": 0,
"capillary_refill_time": 3.3,
"shock_index": 1.32,
"hypoxia_flag": 1,
"hypotension_flag": 0,
"tachycardia_flag": 1,
"severity": 3,
"survival_probability": 50,
"priority_score": 0.9
}

🧠 KEY MODEL INSIGHTS

🔴 Critical Respiratory Cases

- SpO2 < 80
- Severe breathing difficulty
- Airway obstruction / collapsed lung

🟡 Moderate

- SpO2: 80–90
- Painful or labored breathing
- Conscious but distressed

🟢 Mild (not shown here)

- SpO2 > 94
- Minimal symptoms

⚠️ IMPORTANT MODEL BEHAVIOR

Your model should learn:

- **Low SpO2 ≠ always bleeding → respiratory cause**
- **High RR + low SpO2 → airway/lung issue**
- **Bluish skin → severe hypoxia (critical signal)**

🔥 DATASET STRENGTH

These cases add:

✔ Non-bleeding critical conditions
✔ Oxygen-driven severity classification
✔ Strong differentiation from trauma cases

🏆 YOU NOW HAVE COVERAGE OF:

- Blast ✅
- Gunshot ✅
- Shock ✅
- Neuro ✅
- Musculoskeletal ✅
- Respiratory ✅

👉 This is now a **multi-system triage dataset (very advanced level)**

🚀 NEXT STEP

If you want to go **next level (winning level)**:

I can:

- ✅ Combine ALL into **1000+ row dataset (CSV)**
- ✅ Add **time-series progression (deterioration/improvement)**
- ✅ Build **ML model (95%+ accuracy possible on synthetic data)**
- ✅ Create **live demo simulation (judges LOVE this)**

Just tell me 👍

Below is a **clinically consistent, ML-ready dataset block (5 rows)** for
🧊🔥 **Environmental Conditions** — these are critical because they're **non-trauma but life-threatening**, driven by temperature, hydration, and oxygen.

🧊🔥 ENVIRONMENTAL CONDITIONS — DATASET (5 SAMPLES)

1️⃣ Heatstroke → High Temp + Unconscious


```
{
  "age": 33,
  "injury_type": "environmental",
  "scenario": "heatstroke",
  "heart_rate": 145,
  "systolic_bp": 85,
  "spo2": 95,
  "respiratory_rate": 30,
  "body_temperature": 41.2,
  "conscious": 0,
  "skin_color": "hot_dry",
  "hydration_level": "low",
  "environment": "desert",
  "blood_loss_ml": 0,
  "capillary_refill_time": 3.8,
  "shock_index": 1.70,
  "hypoxia_flag": 0,
  "hypotension_flag": 1,
  "tachycardia_flag": 1,
  "severity": 3,
  "survival_probability": 35,
  "priority_score": 0.95
}
```

2️⃣ Hypothermia → Low Temp + Slow Pulse


```
{
  "age": 40,
  "injury_type": "environmental",
  "scenario": "hypothermia",
  "heart_rate": 48,
  "systolic_bp": 90,
  "spo2": 92,
  "respiratory_rate": 12,
  "body_temperature": 34.0,
  "conscious": 1,
  "skin_color": "pale",
  "hydration_level": "normal",
  "environment": "cold",
  "blood_loss_ml": 0,
  "capillary_refill_time": 3.5,
  "shock_index": 0.53,
  "hypoxia_flag": 0,
  "hypotension_flag": 0,
  "tachycardia_flag": 0,
  "severity": 2,
  "survival_probability": 65,
  "priority_score": 0.7
}
```

3️⃣ Dehydration → Weakness + Dizziness


```
{
  "age": 28,
  "injury_type": "environmental",
  "scenario": "dehydration",
  "heart_rate": 110,
  "systolic_bp": 95,
  "spo2": 97,
  "respiratory_rate": 20,
  "body_temperature": 37.8,
  "conscious": 1,
  "mental_state": "dizzy",
  "hydration_level": "low",
  "environment": "desert",
  "blood_loss_ml": 0,
  "capillary_refill_time": 3.0,
  "shock_index": 1.16,
  "hypoxia_flag": 0,
  "hypotension_flag": 0,
  "severity": 1,
  "survival_probability": 85,
  "priority_score": 0.4
}
```

4 High-Altitude Sickness → Low Oxygen


```
{
  "age": 35,
  "injury_type": "environmental",
  "scenario": "high_altitude_sickness",
  "heart_rate": 120,
  "systolic_bp": 100,
  "spo2": 75,
  "respiratory_rate": 28,
  "body_temperature": 36.5,
  "conscious": 1,
  "mental_state": "confused",
  "environment": "high_altitude",
  "hydration_level": "normal",
  "blood_loss_ml": 0,
  "capillary_refill_time": 2.9,
  "shock_index": 1.20,
  "hypoxia_flag": 1,
  "hypotension_flag": 0,
  "severity": 2,
  "survival_probability": 60,
  "priority_score": 0.75
}
```


---

5 Fatigue Collapse → Exhaustion + Low Vitals


```
{
  "age": 30,
  "injury_type": "environmental",
  "scenario": "fatigue_collapse",
  "heart_rate": 85,
  "systolic_bp": 88,
  "spo2": 94,
  "respiratory_rate": 16,
  "body_temperature": 36.8,
  "conscious": 1,
  "mental_state": "weak",
  "hydration_level": "low",
  "environment": "field",
  "blood_loss_ml": 0,
  "capillary_refill_time": 3.2,
  "shock_index": 0.96,
  "hypoxia_flag": 0,
  "hypotension_flag": 1,
  "severity": 1,
  "survival_probability": 80,
  "priority_score": 0.5
}
```

🧠 KEY MODEL INSIGHTS

🔴 Critical Environmental

- Very high temp (>40°C) → heatstroke
- Unconscious + dehydration + tachycardia

🟡 Moderate

- Hypothermia (low HR + low temp)
- High-altitude hypoxia

🟢 Mild

- Dehydration
- Fatigue

⚠️ IMPORTANT MODEL LEARNING

Your model should distinguish:

- **Normal SpO2 + high temp → heat issue (not respiratory)**
- **Low HR + low temp → hypothermia (not neuro)**
- **Low SpO2 + altitude → environmental, not lung damage**

👉 This prevents **misclassification errors**

🔥 DATASET STRENGTH

These cases add:

✔ Non-injury emergencies
✔ Temperature-driven conditions
✔ Environmental awareness
✔ Real-world battlefield realism

🏆 FINAL STATUS OF YOUR DATASET

You now have:

✔ Blast
✔ Gunshot
✔ Shock
✔ Neuro
✔ Musculoskeletal
✔ Respiratory
✔ Environmental

👉 This is now a **complete multi-domain battlefield triage dataset**

🚀 NEXT STEP (HIGHLY RECOMMENDED)

Now you should:

1. Merge all datasets → 500–2000 rows

2. Train model:

- Classification (severity)
- Regression (survival)

3. Build demo:

- Input → prediction → action

If you want:
I can now:

- ✅ Generate **FULL dataset CSV (1000+ rows)**
- ✅ Provide **Python generator script**
- ✅ Build **ML model + API**
- ✅ Create **winning PPT + demo flow**

Just tell me 👍